

Proof: Variance of parameter estimates for simple linear regression

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Theorem: Assume a [simple linear regression model](#) with independent observations

$$y = \beta_0 + \beta_1 x + \varepsilon, \varepsilon_i \sim \mathcal{N}(0, \sigma^2), i = 1, \dots, n \tag{1}$$

and consider estimation using [ordinary least squares](#). Then, the variances of the estimated parameters are

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \frac{x^T x}{n} \cdot \frac{\sigma^2}{(n-1)s_x^2} \\ \text{Var}(\hat{\beta}_1) &= \frac{\sigma^2}{(n-1)s_x^2} \end{aligned} \tag{2}$$

where s_x^2 is the [sample variance](#) of x and $x^T x$ is the sum of squared values of the covariate.

Proof: According to the simple linear regression model in [\(1\)](#), the variance of a single data point is

$$\text{Var}(y_i) = \text{Var}(\varepsilon_i) = \sigma^2. \tag{3}$$

The [ordinary least squares estimates](#) for [simple linear regression](#) are given by

$$\begin{aligned} \hat{\beta}_0 &= \frac{1}{n} \sum_{i=1}^n y_i - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n x_i \\ \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{aligned} \tag{4}$$

If we define the following quantity

$$c_i = \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2}, \tag{5}$$

we note that

$$\begin{aligned} \sum_{i=1}^n c_i^2 &= \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} \right)^2 \\ &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{[\sum_{i=1}^n (x_i - \bar{x})^2]^2} \\ &= \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2}. \end{aligned} \tag{6}$$

With [\(5\)](#), the estimate for the slope from [\(4\)](#) becomes

$$\begin{aligned} \hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \sum_{i=1}^n c_i (y_i - \bar{y}) \\ &= \sum_{i=1}^n c_i y_i - \bar{y} \sum_{i=1}^n c_i \end{aligned} \tag{7}$$

and with [\(3\)](#) and [\(6\)](#) as well as [invariance](#), [scaling](#) and [additivity](#) of the variance, the variance of $\hat{\beta}_1$ is:

$$\begin{aligned} \text{Var}(\hat{\beta}_1) &= \text{Var} \left(\sum_{i=1}^n c_i y_i - \bar{y} \sum_{i=1}^n c_i \right) \\ &= \text{Var} \left(\sum_{i=1}^n c_i y_i \right) \\ &= \sum_{i=1}^n c_i^2 \text{Var}(y_i) \\ &= \sigma^2 \sum_{i=1}^n c_i^2 \\ &= \sigma^2 \frac{1}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{(n-1) \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}{\sigma^2} \\ &= \frac{\sigma^2}{(n-1)s_x^2}. \end{aligned} \tag{8}$$

Finally, with [\(3\)](#) and [\(8\)](#), the variance of the intercept estimate from [\(4\)](#) becomes:

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n y_i - \hat{\beta}_1 \cdot \frac{1}{n} \sum_{i=1}^n x_i \right) \\ &= \text{Var} \left(\frac{1}{n} \sum_{i=1}^n y_i \right) + \text{Var} \left(\hat{\beta}_1 \cdot \bar{x} \right) \\ &= \left(\frac{1}{n} \right)^2 \sum_{i=1}^n \text{Var}(y_i) + \bar{x}^2 \cdot \text{Var}(\hat{\beta}_1) \\ &= \frac{1}{n^2} \sum_{i=1}^n \sigma^2 + \bar{x}^2 \cdot \frac{\sigma^2}{(n-1)s_x^2} \\ &= \frac{\sigma^2}{n} + \frac{\sigma^2 \bar{x}^2}{(n-1)s_x^2}. \end{aligned} \tag{9}$$

Applying the formula for the sample variance s_x^2 , we finally get:

$$\begin{aligned} \text{Var}(\hat{\beta}_0) &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \sigma^2 \left(\frac{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} + \frac{\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \sigma^2 \left(\frac{\frac{1}{n} \sum_{i=1}^n (x_i^2 - 2\bar{x}x_i + \bar{x}^2) + \bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \sigma^2 \left(\frac{\left(\frac{1}{n} \sum_{i=1}^n x_i^2 - 2\bar{x} \frac{1}{n} \sum_{i=1}^n x_i + \bar{x}^2 \right) + \bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \sigma^2 \left(\frac{\frac{1}{n} \sum_{i=1}^n x_i^2 - 2\bar{x}^2 + 2\bar{x}^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \sigma^2 \left(\frac{\frac{1}{n} \sum_{i=1}^n x_i^2}{(n-1) \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2} \right) \\ &= \frac{x^T x}{n} \cdot \frac{\sigma^2}{(n-1)s_x^2}. \end{aligned} \tag{10}$$

■

Sources:

- Penny, William (2006): "Finding the uncertainty in estimating the slope"; in: *Mathematics for Brain Imaging*, ch. 1.2.4, pp. 18-20, eq. 1.37; URL: https://ueapsylabs.co.uk/sites/wpenny/mbi/mbi_course.pdf.
- Wikipedia (2021): "Proofs involving ordinary least squares"; in: *Wikipedia, the free encyclopedia*, retrieved on 2021-10-27; URL: https://en.wikipedia.org/wiki/Proofs_involving_ordinary_least_squares#Unbiasedness_and_variance_of_%7F%22%60UNIQ-postMath-00000037-QINU%60%22%7F.

Metadata: ID: P273 | shortcut: slr-olsvar | author: JoramSoch | date: 2021-10-27, 11:53.

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